RESEARCH PAPER | SUPPLY CHAIN ANALYTICS

Factory Reallocation and Shipping Optimization

A study of Nassau Candy Distributor

Gopal Krushna Panda CUTM, B.Tech Computer Science and Engineering | Unified Mentor Internship Project

ABSTRACT

Nassau Candy Distributor assigns each product to a factory using fixed rules. This study asks whether moving selected products to different factories could shorten shipping time without hurting profit. Using 10,194 orders, three regression models were compared. Gradient Boosting performed best (RMSE 259.97) but explained only a small share of lead-time variation ($R^2 = 0.044$). The 45 shipping routes were then grouped into three clusters, and 24 reassignment scenarios were simulated. The results point to possible operational gains, but the lead-time calculation needs to be verified before any of the findings are relied on.

10,194	45	24	0.044
orders analysed	routes clustered	scenarios simulated	best model R^2

1. Introduction

Nassau Candy operates five factories, located in Arizona, Georgia, Minnesota, Illinois and Tennessee. Every product is tied to a single factory. Some regions see longer lead times, and shipping costs can squeeze margins. Before changing any factory assignment, the company needs evidence on which routes underperform and whether a change would actually help.

The study is organised around three questions:

- Which factors are associated with shipping lead time?
- Do some routes perform clearly worse than others?
- Which product reassignments look promising, and what might they cost in profit?

2. Data

The dataset holds 10,194 orders across 18 columns, with no missing values. It spans 15 products, 4 regions and 4 shipping modes. Chocolate makes up most of the orders (96.5%), followed by Other (3%) and Sugar (0.4%). Each product maps to one factory. Wonka Bars from Lot’s O’ Nuts, for example, are made in Arizona, while Sugar Shack in Minnesota produces Laffy Taffy, SweeTARTS, Nerds and Fun Dip.

Measure	Value
Orders	10,194
Columns	18 (no missing values)
Products / regions / shipping modes	15 / 4 / 4
Division share of orders	Chocolate 96.5%, Other 3.0%, Sugar 0.4%
Total gross profit	$93,442.80
Reported margin	65.9%

Table 1. Dataset summary.

3. Method

3.1 Data preparation

Dates were parsed as DD-MM-YYYY and lead time was calculated from them. Additional features were profit margin, order month, day of week and units per sale. Region, shipping mode and division were converted to numeric codes with LabelEncoder.

3.2 Regression models

The data was split 80/20 into 8,155 training rows and 2,039 test rows. Linear Regression, Random Forest and Gradient Boosting were compared on RMSE, MAE and R^2.

Model	RMSE	MAE	R^2
Linear Regression	263.68	212.40	0.017
Random Forest	263.43	216.94	0.019
Gradient Boosting	259.97	213.64	0.044

Table 2. Test-set performance. Gradient Boosting (highlighted) was selected.

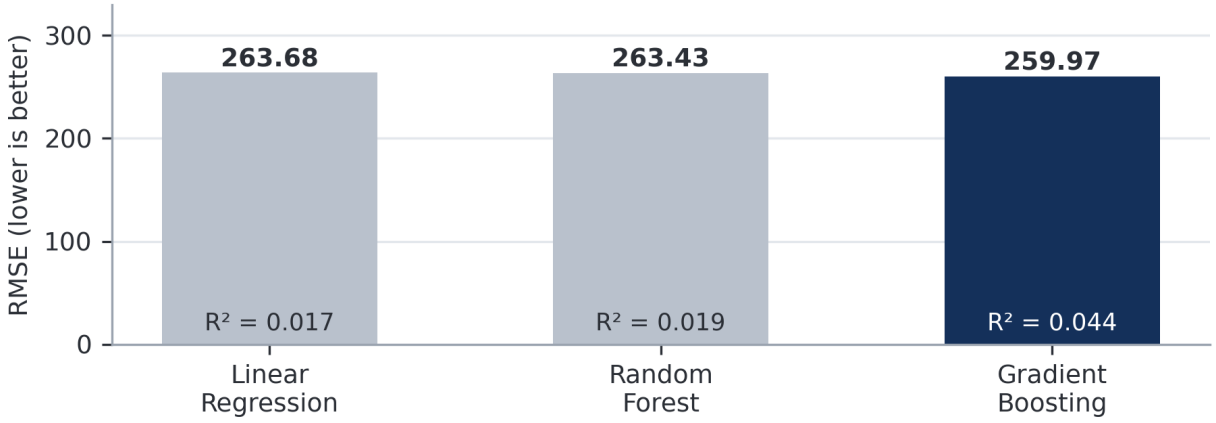


Figure 1. RMSE by model, with R^2 shown inside each bar.

Gradient Boosting had the lowest RMSE and was used for the later estimates. All three R^2 values are low, which means the chosen features explain very little of the variation in lead time. An error of roughly 260 days is also far larger than any realistic shipping time, which supports the concern raised in Section 5 about how lead time was computed.

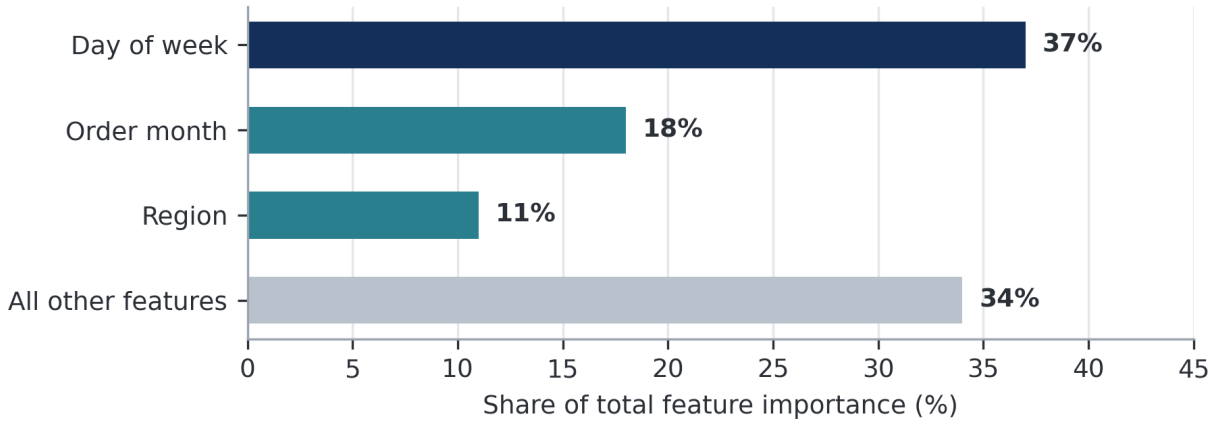


Figure 2. Feature importance reported for the Gradient Boosting model.

Day of week carried the most importance (37%), followed by order month (18%) and region (11%). Calendar variables outranking region is unusual for a shipping problem, and is another reason to recheck the date handling.

3.3 Route clustering

A route was defined as a combination of region, shipping mode and division, giving 45 routes. K-Means with k = 3 produced a silhouette score of 0.406.

3.4 Reassignment scenarios

For each selected product, a move to every alternative factory was simulated, and the estimated lead time and profit were compared with the current assignment. In total, 24 scenarios were evaluated across six products.

4. Results

4.1 Route clusters



Figure 3. Distribution of the 45 routes across the three clusters.

The clustering produced 29 fast routes, 10 medium routes and 6 slow routes. All six slow routes involved Sugar products. However, the slow group covers only 7 orders, so this is a weak signal and should be read as a lead for further checking, not a conclusion.

4.2 Reassignment scenarios

The two strongest individual scenarios each showed an estimated saving of 25.5 days, and the Gobstopper move also showed a $0.68 profit gain. The five leading scenarios together were reported to save 127.1 days, add $2.72 in profit and touch 1,361 orders. Against total gross profit of $93,442.80, that profit gain is very small, and these figures have not yet been independently checked.

Scenario	Estimated days saved	Estimated profit change	Orders affected
Fun Dip to Secret Factory	25.5	Not reported	Not reported
Everlasting Gobstopper to Lot's O' Nuts	25.5	+$0.68	Not reported
Top five scenarios combined	127.1	+$2.72	1,361

Table 3. Leading reassignment scenarios (simulated).

4.3 Risk score

The top recommendations received a risk score of 49.4 out of 100, labelled low. A score near the middle of a 100-point scale is an odd fit for a "low" label, so the scoring method and the band definitions should be documented before this number is used.

5. Discussion

The regression results are weak, and the date issue may be undermining the whole analysis. The Sugar routes are worth a closer look because they sit in the slow cluster, but the cluster holds very few orders and cannot be treated as conclusive. The reassignment results are simulations, not real-world tests, so the scenarios are best read as hypotheses to validate rather than instructions to move production.

Any real decision would also need to weigh factory capacity, production costs, service levels, inventory and other operational limits.

6. Limitations

- The calculated lead-time values may be incorrect.
- The regression models have low explanatory power.
- The slow cluster rests on a very small number of orders.
- Reassignment scenarios were simulated and have not been tested in operations.
- The data comes from a single distributor, so the findings may not generalise to other businesses.

7. Conclusion and Next Steps

This work offers an initial framework for examining factory allocation and shipping performance at Nassau Candy Distributor. Gradient Boosting was the best of the three models, but an R^2 of 0.044 is too low to support confident lead-time predictions. The clustering and reassignment simulations suggest directions for further analysis. They are not proven improvements.

> **NEXT STEPS**
> First, confirm the date format and the lead-time calculation, then rerun the analysis. Second, check the reassignment estimates against actual transit times, factory capacity, production costs and service constraints before anything is recommended for implementation.